

Supplementary material

Semantic Identifiability under Band-Order Ambiguity in PlanetScope Landslide Segmentation

Supplementary Method S1: documented CAS external check

The executed code assigns eight CAS RGB subdatasets by a salted hash of the region name. The only surviving protocol was corrected and first committed after execution; it is therefore a post-execution record, not independently verifiable prospective specification or preregistration. Development used Lombok, Moxitaidi UAV-1m, Hokkaido, and Wenchuan; Palu was the sole validation region; Mengdong, Moxi town UAV-1m, and Tiburon Peninsula PlanetScope were designated evaluation regions. Wenchuan replaced Moxi town in development after one loader-semantics access, before model fitting; neither predictions nor metrics were inspected. The same three-channel zero-control residual-attention model was trained with base or plain boundary loss for seeds 42, 43, and 44. Fixed non-overlapping 128-pixel crops, optimization budget, augmentations, and validation-only threshold grid were shared. The executed limits were 80 source tiles per development region and 160 per validation/test region, each yielding 16 crops. The post-run record lists four quantitative conditions: at least +1.5 boundary-F1 points, positive effects in all three held regions, F1 non-inferiority within one point, and non-negative F1 effects for at least two seeds. Independent recomputation verified all 150 event metrics. The observed event-macro changes were -2.19 F1 points and -1.86 boundary-F1 points; held-region boundary effects were -0.39 , -5.14 , and -0.05 points.

Table S1: Paired plain-boundary minus base-loss effects by designated CAS region and seed, in F1/boundary-F1 percentage points. The macro row averages the three regions within each seed.

Region	Seed 42	Seed 43	Seed 44
Mengdong	$-8.66 / +1.54$	$-2.40 / -1.57$	$+1.74 / -1.15$
Moxi town	$-12.73 / -1.16$	$+3.50 / -10.69$	$+0.13 / -3.58$
Tiburon Peninsula	$-9.88 / -0.38$	$+2.57 / -0.24$	$+5.99 / +0.46$
Event macro	$-10.42 / +0.00$	$+1.22 / -4.17$	$+2.62 / -1.42$

Only one of four quantitative conditions passed: two seeds had non-negative macro F1 effects. The mean boundary-F1 gain was below +1.5 points, not every held region had a positive mean boundary effect, and mean F1 was below the -1 -point non-inferiority bound. A separate process condition required no post-access code, threshold, or reporting-rule change; because the protocol was first committed after execution, that condition is not independently time-verifiable and is not counted as passed.

Supplementary Method S2: internally precommitted LRD stress test

The protocol, code, 16-development/6-validation/6-protected EID split, crop rule, outcomes, and four criteria were locally committed as 97b107836a3737436b973adb28d0f5c5be6c3d63 before validation or protected optical access, but no public third-party pre-access timestamp exists. Each EID uses up to 64 coordinate-hashed 128-pixel crops from named POST1 Sentinel-2 B04/B03/B02 rasters and its mask. Six zero-control models cross base/plain-boundary loss with seeds 42–44. Checkpoints and validation thresholds were hashed before a protected-fetch authorization was issued. Independent recomputation matched all 330 metrics.

All four frozen criteria fail, but validation maximum F1 is 0.008–0.151 and 84.1% of protected crops are empty. The frozen boundary effect is -24.10 points; empty/empty scoring of zero gives -0.15 , pooled scoring -4.92 , and a common 0.5 threshold -0.55 . A family audit groups development PH0003 with protected PH0001/PH0004, leaving four non-overlapping EIDs. Accordingly the run is failed/indeterminate sensitivity evidence, not confirmation.

Table S2: Frozen LRD plain-boundary minus base-loss effects. EID rows average seeds; seed rows average six protected EIDs.

Unit	$\Delta F1$ (points)	$\Delta BF1$ (points)
CA0002	-5.42	-25.56
CN0002	$+0.66$	-22.38
IR0002	$+0.22$	-25.77
PH0001	-10.10	-24.78
PH0004	-10.57	-21.45
US0001	-3.18	-24.64
Seed 42 macro	$+0.98$	-77.19
Seed 43 macro	-0.04	-0.06
Seed 44 macro	-15.13	$+4.96$
Across-seed macro	-4.73	-24.10

Supplementary Method S3: authorized Sen12 stress test

Sen12Landslides provides named Sentinel-2 B02/B03/B04/B08 variables and inventory labels. The public protocol evolved fail-closed before model outcomes: v1 omitted official task membership; v2 exposed the `usa_puertorico`→`usa` alias; v3 failed its China-held-out development floor; v4 passed scientific floors but failed an over-strict float32 reproduction check; and v5 was publicly locked (`a4fc497`), fitted, and authorized (`3f39077`) before 1,889 selected files from ten protected inventories were extracted. Version 6 was a public post-access execution correction: one Indonesia post-image SCL array contained 255, outside official classes 0–11. The correction retains 255 in histograms and excludes it only from descriptive cloud denominators; no file, input, mask, endpoint, or condition changed. It was published before model inference. One protected MASK count observed during diagnosis is disclosed.

Independent verification recomputed 420 primary metric cells, 294 high-confidence cells, all ten SCL summaries, and every registered condition. Only condition 2 passed. The registered transformed-envelope check reports one Kyrgyzstan overlap, whereas the native same-CRS pixel-edge rectangles only share an edge and have zero positive-area overlap. The former fails the frozen bar; the latter prevents interpreting that failure as demonstrated duplication.

Supplementary Table S4 reports the detailed RGBN-candidate HR-GLDD operating metrics underlying the compact main comparison.

Table S3: Registered Sen12 cross-dataset recurrence conditions. Intervals are inventory-bootstrap 95% intervals. Contrast values are percentage points.

Condition	Estimate		Registered requirement	Result
NDVI attention minus raw edge F1	+0.70	[+0.28, +1.16]	95% upper endpoint < +1.00	Fail
NDVI-modulated minus plain-boundary F1	pt		pt	
Generic plain-boundary F1	−0.06	[−0.41, +0.26]	95% upper endpoint < +1.00	Pass
	pt		pt	
	+0.61	[−0.09, +1.32]	All control means > 0; pooled lower endpoint > 0; at least 8/10 positive	Fail
Absolute transfer	Arm means 1.21–3.14% F1	1.21–3.14% F1	Every arm mean at least 10% F1	Fail
Registered spatial integrity	1 transformed-envelope overlap	EPSG:4326 transformed-envelope overlap	Zero cross-inventory envelope overlaps	Fail

Table S4: Detailed RGBN-candidate HR-GLDD outcomes. Values are three-seed mean (population standard deviation); thresholds are validation-selected.

Configuration	F1 (%)	Boundary F1 (%)	Default FPR (%)
U-Net/base	66.92 (0.68)	49.62 (0.63)	2.19 (0.06)
ResU-Net/base	68.55 (0.56)	52.93 (0.79)	2.30 (0.08)
Attn-zero/base	68.88 (0.58)	53.17 (0.25)	2.23 (0.03)
Attn-raw/base	69.05 (0.25)	52.92 (0.32)	2.25 (0.04)
Attn-index/base	69.04 (0.27)	52.76 (0.33)	2.24 (0.04)
Attn-index/plain-boundary	69.98 (0.56)	55.96 (0.45)	2.49 (0.09)
Attn-zero/plain-boundary	69.99 (0.52)	56.04 (0.43)	2.50 (0.08)
Attn-raw/plain-boundary	69.88 (0.53)	55.85 (0.73)	2.49 (0.14)
Attn-index/index-boundary	69.58 (1.02)	56.34 (0.52)	2.47 (0.12)

Configuration	Matched FPR (%)	Matched precision (%)	10-m F1 (%)
U-Net/base	2.19 (0.06)	76.86 (0.55)	61.84 (0.42)
ResU-Net/base	1.99 (0.06)	78.27 (0.20)	64.22 (0.37)
Attn-zero/base	1.98 (0.09)	78.59 (0.56)	63.71 (0.45)
Attn-raw/base	2.00 (0.07)	78.57 (0.36)	63.92 (0.14)
Attn-index/base	1.99 (0.09)	78.63 (0.56)	63.93 (0.25)
Attn-index/plain-boundary	2.00 (0.08)	78.49 (0.40)	65.14 (0.44)
Attn-zero/plain-boundary	1.97 (0.05)	78.63 (0.16)	65.35 (0.47)
Attn-raw/plain-boundary	2.00 (0.10)	78.49 (0.54)	65.37 (0.30)
Attn-index/index-boundary	1.98 (0.07)	78.54 (0.54)	65.06 (0.51)

Table S5: Two-order sensitivity envelopes over release-supported RGBN and conservative native-order BGRN sensitivity. Envelopes span mapping-specific descriptive means without assigning equal plausibility; they are not confidence intervals.

Contrast (points)	RGBN	BGRN	Envelope	Direction
Index input vs raw, F1	−0.01	−0.21	[−0.21, −0.01]	negative
Index vs plain loss, F1	−0.41	−0.33	[−0.41, −0.33]	negative
Index vs plain loss, BF1	+0.38	−0.05	[−0.05, +0.38]	unidentified
Plain boundary vs base, F1	+0.96	+1.13	[+0.96, +1.13]	positive

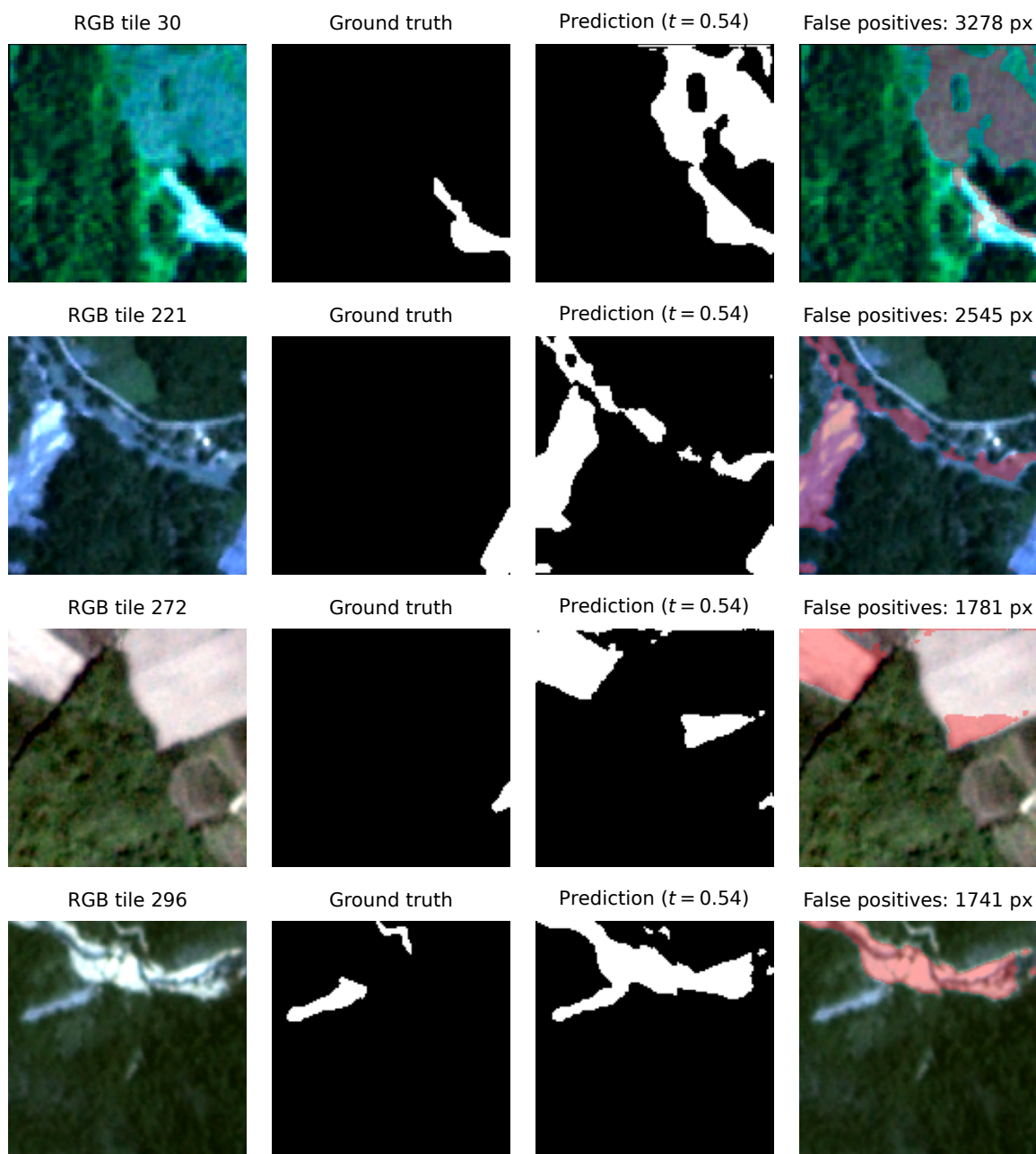


Figure S1: Four low-prevalence HR-GLDD test tiles with the largest false-positive counts for zero-control plus plain-boundary weighting at validation-matched recall (seed 42). Visible composites follow the official notebook’s direct columns-0–2 display and are not band-order evidence. Red overlays denote false-positive pixels; background categories are not annotated.